

3.2.2 Biomechanical movement

Students should develop knowledge and understanding of motion and forces, and their relevance to performance in physical activity and sport.

Students should have a knowledge and use of biomechanical definitions, equations, formulae and units of measurement and demonstrate the ability to plot, label and interpret biomechanical graphs and diagrams.

3.2.2.1 Biomechanical principles

Content	Additional information
Newton's Three Laws of linear motion applied to sporting movements.	First law (inertia), second law (acceleration), third law (action/reaction). Force.
Definitions, equations and units of example scalars.	Speed, distance.
Centre of mass.	
Factors affecting stability.	Height of centre of mass, area of base of support, position of line of gravity and body mass.

3.2.2.2 Levers

Content	Additional information
Three classes of lever and examples of their use in the body during physical activity and sport.	
Mechanical advantage and mechanical disadvantage of each class of lever.	

3.2.2.3 Linear motion

Content	Additional information
An understanding of the forces acting on a performer during linear motion.	Gravity, frictional force, air resistance, internal-muscular force, weight.
Definitions, equations and units of vectors.	Weight, velocity, displacement, acceleration and momentum.
Definitions, equations and units of scalars.	Mass, speed and distance.
The relationship between impulse and increasing and decreasing momentum in sprinting through the interpretation of force/time graphs.	

3.2.2.4 Angular motion

Content	Additional information
Application of Newton's laws to angular motion.	
Definitions and units for angular motion.	Angular displacement, angular velocity, angular acceleration.

Content	Additional information
Conservation of angular momentum during flight, moment of inertia and its relationship with angular velocity.	

3.2.2.5 Projectile motion

Content	Additional information
Factors affecting horizontal displacement of projectiles.	
Factors affecting flight paths of different projectiles.	Shot put, badminton shuttle.
Vector components of parabolic flight.	

3.2.2.6 Fluid mechanics

Content	Additional information
Dynamic fluid force.	Drag and lift.
Factors that reduce and increase drag and their application to sporting situations.	
The Bernoulli principle applied to sporting situations.	Upward lift force (discus). Downward lift force (speed skiers, cyclists, racing cars).